Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/04/2025
PI Name:	Kiffer, Frederico C. Ph.D.		
Project Title:	Effects of Galactic Cosmic Radiation on Translationally-Relevant Cognitive Behaviors and Response to Social Stress		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-4318	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 TRISH RFA-1901-PD Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	08/01/2019	**End Date:**	07/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2022 per E. Urquieta/TRISH (Ed., 7/1/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-P0402		
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Task Description:	POSTDOCTORAL FELLOWSHIP Astronauts on a Mars mission will be exposed to potentially harmful levels of charged-particle radiation. Thirty years of basic research with ground-based charged-particle radiation has provided overwhelming evidence that the rodent Central Nervous System (CNS) and behavior are negatively affected by charged-particles, suggesting this is a concern for astronaut health. However, the current literature has several caveats, including using young rather than fully-adult rodents, lack of studies focused on the female rodent CNS, unrealistic single-particle radiation simulations, lack of feasible therapeutic countermeasures, and behavioral tests with low translational potential and high handling- or experimenter-induced variability. Additionally, no published CNS study has yet assessed the effects of charged-particle radiation in combination with spaceflight-relevant stressors. We will address these limitations by exposing male and female mice of astronaut age to a novel, complex, but realistic Galactic Cosmic Ray simulation (GCR) then examining network-dependent behavioral performance on a highly-translational, appetitive touch-screen platform, including a behavioral test battery that can be viewed as analogous to the tests currently used on astronauts aboard the International Space Station. A separate group of mice will be examined for the influence of GCR on their ability to cope with social and physical stress in a validated mouse model of depression. Finally, we will test a promising anti-inflammatory drug and its ability to prevent GCR-induced deficits in cognition and social defeat stress. This proposal will address a wide range of NASA-defined knowledge gaps in its Human Research Roadmap. Specifically, we strive to provide a model basis for identifying individuals who are resilient to the extreme spaceflight conditions, validate a measure for monitoring behavioral health, determine radiation dose thresholds for behavioral measures in a domain basis and in combination with social defeat stress, all of which will help inform NASA's risk models for a crewed mission to Mars.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have demonstrated that GCR remains a pressing concern for the behavioral health of mice due to observed behavioral deficits following exposure in male and female mice. These observations in a mouse model suggest that astronauts on a mission to Mars, who are expected to receive similar doses of GCR might be at risk of similar behavioral endpoints. Our work also demonstrates that CDDO-EA might be a promising dietary countermeasure to prevent radiation-induced cognitive deficits in female mice. However, CDDO-EA was not effective in preventing radiation-induced social memory deficits in male mice. In both male and female mice, CDDO-EA in combination with ground-based GCR exposure resulted in sociability deficits as compared to control mice. Further inquiry into CDDO-EA is warranted. [Ed. Note: CDDO-EA is a chemical compound. For more information about CDDO-EA, visit the National Library of Medicine website: https:// .
Task Progress:	Our preliminary data, which result from one of the first rodent exposures to the NASA Space Radiation Laboratory (NSRL's) acute 33-beam galactic cosmic ray (GCR), suggest that the mission-relevant dose of 750mGy does not result in the same behavioral deficits commonly reported in the monoenergetic single- or simple ion combination literature for aged mice. However, we still observed radiation-induced deficits following radiation exposure. We show that this novel radiation paradigm carries cognitive behavioral risk in mice, suggesting the possibility that the same may be true in human exposures. We also demonstrate radio-protective properties of CDDO-EA and evaluate its use as a candidate radiation countermeasure. CDDO-EA prevented numerous radiation-related deficits in females, but did not appear to be an effective countermeasure in males.
Bibliography Type:	Description: (Last Updated: 11/17/2025)
Articles in Peer-reviewed Journals	Yun S, Kiffer FC, Bancroft GL, Guzman CS, Soler I, Haas HA, Shi R, Patel R, Lara-Jiménez J, Kumar PL, Tran FH, Ahn KJ, Rong Y, Luitel K, Shay JW, Eisch AJ. "The longitudinal behavioral effects of acute exposure to galactic cosmic radiation in female C57BL/6J mice: Implications for deep space missions, female crews, and potential antioxidant countermeasures." J Neurochem. 2025 Jan;169(1):e16225. https://doi.org/10.1111/jnc.16225 ; PubMed PMID: 39318241; PubMed Central PMCID: PMC11658192 , Jan-2025